CROW CARRIER — eight synchronous microphone channels / USB data / 12 V operating power

PROTOTYPE — NOT FOR FABRICATION. Outdoor protection and exact assembly qualification remain open.

12 V INPUT / REVERSE PROTECTION / 5 V CONVERSION

J1
VIN12 1
GND 2
CENTER_12V
SLEEVE_GND
PJ-102AH / 12V CENTER POSITIVE

F1
VIN12 1
2
F1650312
0451001.MRL 1A

D3
2
1
SS34-E3/57T

12V

D1
LittleFuse SMBJ15A
GND

C1
10uF 35V X7R 20%
GND

U10
VIN 1
GND 2
VOUT 3
R-78E5.0-0.5

5V

D2
2
1
VSYS
Diodes Inc 1N5819HW-7-F

12V
C2
10uF 35V X7R 20%
GND

PWR1
VIN12
PWR_FLAG

PWR2
12V
PWR_FLAG

PWR3
GND
PWR_FLAG

PWR4
VSYS
PWR_FLAG

ADC 1: CHANNELS 1-4 / PCM1864 / common 3V3

U1
ADC1_CH1_M 1 VIN1M
ADC1_CH2_M 2 VIN2M
ADC1_CH1_P 3 VIN1P
ADC1_CH2_P 4 VIN2P
REF1 6 MIC_BIAS
GND 7 VREF
3V3 8 AGND
AI
AO
LDO1 11 LDO
GND 12 DGND
3V3 13 DVDD
3V3 14 DVDD
MCLK_ADC1 15 SCKI
LRCK 16
BCK 17
DOUT 18
GPO0 19
GPO1 20
MISO 21
SDA 22
SCL 23
MCLK 24
GND 25
MOSI 26
GND 27
VINL4 28 ADC1_CH4_M
VINR4 29 ADC1_CH4_P
VINL3 30 ADC1_CH3_M
VINR3 31 ADC1_CH3_P
PCM1864DBT

U1 LOCAL DECOUPLING — separate capacitor positions at each supply pin

REF1
C11
1uF 10V X7R VREF
GND

LDO1
C12
100nF 10V X7R LDO
GND

3V3
C13
100nF 10V X7R AVDD
GND

3V3
C14
10uF 10V X7R AVDD
GND

3V3
C15
100nF 10V X7R DVDD
GND

3V3
C16
10uF 10V X7R DVDD
GND

LDO1
C17
10uF 10V X7R LDO
GND

3V3
C18
100nF 10V X7R IOVDD
GND

ADC 2: CHANNELS 5-8 / PCM1864 / common 3V3

U2
ADC2_CH1_M 1 VIN1M
ADC2_CH2_M 2 VIN2M
ADC2_CH1_P 3 VIN1P
ADC2_CH2_P 4 VIN2P
REF2 6 MIC_BIAS
GND 7 VREF
3V3 8 AGND
AI
AO
LDO2 11 LDO
GND 12 DGND
3V3 13 DVDD
3V3 14 DVDD
MCLK_ADC2 15 SCKI
LRCK 16
BCK 17
DOUT 18
GPO0 19
GPO1 20
MISO 21
SDA 22
SCL 23
MCLK 24
GND 25
MOSI 26
GND 27
VINL4 28 ADC2_CH4_M
VINR4 29 ADC2_CH4_P
VINL3 30 ADC2_CH3_M
VINR3 31 ADC2_CH3_P
PCM1864DBT

U2 LOCAL DECOUPLING — separate capacitor positions at each supply pin

REF2
C21
1uF 10V X7R VREF
GND

LDO2
C22
100nF 10V X7R LDO
GND

3V3
C23
100nF 10V X7R AVDD
GND

3V3
C24
10uF 10V X7R AVDD
GND

3V3
C25
100nF 10V X7R DVDD
GND

3V3
C26
10uF 10V X7R DVDD
GND

LDO2
C27
10uF 10V X7R LDO
GND

3V3
C28
100nF 10V X7R IOVDD
GND

U3 ASSEMBLY: DEPOPULATE ON-MODULE D1 only: retain R10/R1 VBUS sense. USB DATA ONLY.

U3
SDA 1 GPO0 24
SCL 2 GPO1 23
GND 3 GND 22 GND
TDM 4 GPO2 21
BCK 5 GPO3 20
LRCK 6 GPO4 19
BUZZ_GPO1 7 GPO5 18
GND 8 GND 17 GND
BUZZ_GPO2 9 GPO6 16
BUZZ_GPO3 10 GPO7 15
BUZZ_GPO4 11 GPO8 14
BUZZ_GPO5 12 GPO9 13
GND 13 GND 12 GND
BUZZ_GPO6 14 GPO10 11
BUZZ_GPO7 15 GPO11 10 ADC_VREF
BUZZ_GPO8 16 GPO12 9 3V3
GPO13 8
GND 17 GND 7 GND
GPO14 6 VSYS
GPO15 5 VBUS
Pico SC0915 MOD: on-module D1 DEPOPULATED

3V3
R1
4.7k 1%
SDA

3V3
R2
4.7k 1%
SCL

SHARED 12.288 MHz MCLK

X1
3V5 1 OE
GND 2 GND
OUT 3
VDD 4
MCLK_SRC
3V3
CB3LV-3I-12M2880000

3V3
C4
100nF 10V X7R
GND

R3
MCLK_SRC 1
2 MCLK_FAN
68R 1% 0.1W 25V

R4
1
2 MCLK_ADC1
56R 1% 0.1W 25V

R5
1
2 MCLK_ADC2
56R 1% 0.1W 25V

POD 1 — differential audio + power + active-low calibration command

J101
P1_A_P 1 contact_1 contact_5 5 GND
P1_A_N 2 contact_2 contact_6 6 GND
P1_12V 3 contact_3 contact_7 7 P1_BUZZ_N
P1_12V 4 contact_4 contact_8 8 GND
RJHSE-5080

D501
P1_A_P 2
GND 1
P1_A_N 3
5 P1_BUZZ_N
SP0504BAHTG

C102
P1_A_P 1
2
2.2uF 25V X7R 20%

R102
1
2 P1_CP
2.7k 1%

ADC1_CH1_P

D102
BZT52C3V9-7-F
GND

C301
2.2nF 50V C0G 5%
GND

C103
P1_A_N 1
2
2.2uF 25V X7R 20%

R103
1
2 P1_CM
2.7k 1%

ADC1_CH1_M

D103
BZT52C3V9-7-F
GND

C401
2.2nF 50V C0G 5%
GND

Q101
P1_BUZZ_N
BUZZ_GPO1
DMN3023L-7

R301
100k 1%
GND

F101
12V 1
2 P1_12V
1206L010/30WR

POD 2 — differential audio + power + active-low calibration command

J102
P2_A_P 1 contact_1 contact_5 5 GND
P2_A_N 2 contact_2 contact_6 6 GND
P2_12V 3 contact_3 contact_7 7 P2_BUZZ_N
P2_12V 4 contact_4 contact_8 8 GND
RJHSE-5080

D502
P2_A_P 2
GND 1
P2_A_N 3
5 P2_BUZZ_N
SP0504BAHTG

C104
P2_A_P 1
2
2.2uF 25V X7R 20%

R104
1
2 P2_CP
2.7k 1%

ADC1_CH2_P

D104
BZT52C3V9-7-F
GND

C302
2.2nF 50V C0G 5%
GND

C105
P2_A_N 1
2
2.2uF 25V X7R 20%

R105
1
2 P2_CM
2.7k 1%

ADC1_CH2_M

D105
BZT52C3V9-7-F
GND

C402
2.2nF 50V C0G 5%
GND

Q102
P2_BUZZ_N
BUZZ_GPO2
DMN3023L-7

R302
100k 1%
GND

F102
12V 1
2 P2_12V
1206L010/30WR

POD 3 — differential audio + power + active-low calibration command

J103
P3_A_P 1 contact_1 contact_5 5 GND
P3_A_N 2 contact_2 contact_6 6 GND
P3_12V 3 contact_3 contact_7 7 P3_BUZZ_N
P3_12V 4 contact_4 contact_8 8 GND
RJHSE-5080

D503
P3_A_P 2
GND 1
P3_A_N 3
5 P3_BUZZ_N
SP0504BAHTG

C106
P3_A_P 1
2
2.2uF 25V X7R 20%

R106
1
2 P3_CP
2.7k 1%

ADC1_CH3_P

D106
BZT52C3V9-7-F
GND

C303
2.2nF 50V C0G 5%
GND

C107
P3_A_N 1
2
2.2uF 25V X7R 20%

R107
1
2 P3_CM
2.7k 1%

ADC1_CH3_M

D107
BZT52C3V9-7-F
GND

C403
2.2nF 50V C0G 5%
GND

Q103
P3_BUZZ_N
BUZZ_GPO3
DMN3023L-7

R303
100k 1%
GND

F103
12V 1
2 P3_12V
1206L010/30WR

POD 4 — differential audio + power + active-low calibration command

J104
P4_A_P 1 contact_1 contact_5 5 GND
P4_A_N 2 contact_2 contact_6 6 GND
P4_12V 3 contact_3 contact_7 7 P4_BUZZ_N
P4_12V 4 contact_4 contact_8 8 GND
RJHSE-5080

D504
P4_A_P 2
GND 1
P4_A_N 3
5 P4_BUZZ_N
SP0504BAHTG

C108
P4_A_P 1
2
2.2uF 25V X7R 20%

R108
1
2 P4_CP
2.7k 1%

ADC1_CH4_P

D108
BZT52C3V9-7-F
GND

C304
2.2nF 50V C0G 5%
GND

C109
P4_A_N 1
2
2.2uF 25V X7R 20%

R109
1
2 P4_CM
2.7k 1%

ADC1_CH4_M

D109
BZT52C3V9-7-F
GND

C404
2.2nF 50V C0G 5%
GND

Q104
P4_BUZZ_N
BUZZ_GPO4
DMN3023L-7

R304
100k 1%
GND

F104
12V 1
2 P4_12V
1206L010/30WR

POD 5 — differential audio + power + active-low calibration command

J105
P5_A_P 1 contact_1 contact_5 5 GND
P5_A_N 2 contact_2 contact_6 6 GND
P5_12V 3 contact_3 contact_7 7 P5_BUZZ_N
P5_12V 4 contact_4 contact_8 8 GND
RJHSE-5080

D505
P5_A_P 2
GND 1
P5_A_N 3
5 P5_BUZZ_N
SP0504BAHTG

C110
P5_A_P 1
2
2.2uF 25V X7R 20%

R110
1
2 P5_CP
2.7k 1%

ADC2_CH1_P

D110
BZT52C3V9-7-F
GND

C305
2.2nF 50V C0G 5%
GND

C111
P5_A_N 1
2
2.2uF 25V X7R 20%

R111
1
2 P5_CM
2.7k 1%

ADC2_CH1_M

D111
BZT52C3V9-7-F
GND

C405
2.2nF 50V C0G 5%
GND

Q105
P5_BUZZ_N
BUZZ_GPO5
DMN3023L-7

R305
100k 1%
GND

F105
12V 1
2 P5_12V
1206L010/30WR

POD 6 — differential audio + power + active-low calibration command

J106
P6_A_P 1 contact_1 contact_5 5 GND
P6_A_N 2 contact_2 contact_6 6 GND
P6_12V 3 contact_3 contact_7 7 P6_BUZZ_N
P6_12V 4 contact_4 contact_8 8 GND
RJHSE-5080

D506
P6_A_P 2
GND 1
P6_A_N 3
5 P6_BUZZ_N
SP0504BAHTG

C112
P6_A_P 1
2
2.2uF 25V X7R 20%

R112
1
2 P6_CP
2.7k 1%

ADC2_CH2_P

D112
BZT52C3V9-7-F
GND

C306
2.2nF 50V C0G 5%
GND

C113
P6_A_N 1
2
2.2uF 25V X7R 20%

R113
1
2 P6_CM
2.7k 1%

ADC2_CH2_M

D113
BZT52C3V9-7-F
GND

C406
2.2nF 50V C0G 5%
GND

Q106
P6_BUZZ_N
BUZZ_GPO6
DMN3023L-7

R306
100k 1%
GND

F106
12V 1
2 P6_12V
1206L010/30WR

POD 7 — differential audio + power + active-low calibration command

J107
P7_A_P 1 contact_1 contact_5 5 GND
P7_A_N 2 contact_2 contact_6 6 GND
P7_12V 3 contact_3 contact_7 7 P7_BUZZ_N
P7_12V 4 contact_4 contact_8 8 GND
RJHSE-5080

D507
P7_A_P 2
GND 1
P7_A_N 3
5 P7_BUZZ_N
SP0504BAHTG

C114
P7_A_P 1
2
2.2uF 25V X7R 20%

R114
1
2 P7_CP
2.7k 1%

ADC2_CH3_P

D114
BZT52C3V9-7-F
GND

C307
2.2nF 50V C0G 5%
GND

C115
P7_A_N 1
2
2.2uF 25V X7R 20%

R115
1
2 P7_CM
2.7k 1%

ADC2_CH3_M

D115
BZT52C3V9-7-F
GND

C407
2.2nF 50V C0G 5%
GND

Q107
P7_BUZZ_N
BUZZ_GPO7
DMN3023L-7

R307
100k 1%
GND

F107
12V 1
2 P7_12V
1206L010/30WR

POD 8 — differential audio + power + active-low calibration command

J108
P8_A_P 1 contact_1 contact_5 5 GND
P8_A_N 2 contact_2 contact_6 6 GND
P8_12V 3 contact_3 contact_7 7 P8_BUZZ_N
P8_12V 4 contact_4 contact_8 8 GND
RJHSE-5080

D508
P8_A_P 2
GND 1
P8_A_N 3
5 P8_BUZZ_N
SP0504BAHTG

C116
P8_A_P 1
2
2.2uF 25V X7R 20%

R116
1
2 P8_CP
2.7k 1%

ADC2_CH4_P

D116
BZT52C3V9-7-F
GND

C308
2.2nF 50V C0G 5%
GND

C117
P8_A_N 1
2
2.2uF 25V X7R 20%

R117
1
2 P8_CM
2.7k 1%

ADC2_CH4_M

D117
BZT52C3V9-7-F
GND

C408
2.2nF 50V C0G 5%
GND

Q108
P8_BUZZ_N
BUZZ_GPO8
DMN3023L-7

R308
100k 1%
GND

F108
12V 1
2 P8_12V
1206L010/30WR

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